

Electric Vehicle Hardware Lab

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

6419

Credits

4

Periods

5/week

Course Details

Title: Electric Vehicle Hardware Lab**Department:** Common**Revision:** REV_Electric Vehicle Hardware Lab**Pattern:** Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 6419 — Electric Vehicle Hardware Lab**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ു൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ English technical terms- ു൬൬൬൬൬൬൬൬൬. ു൬ ു൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬, formula, diagram, units, definitions ു൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬൬൬൬ ു൬൬൬൬.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Electric Vehicle Hardware Lab.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ഐക്യരൂപം സാങ്കേതികവിദ്യയിൽ ഉപയോഗിക്കുന്നതിനായി സാങ്കേതികവിദ്യയെ പരിശോധിക്കുക. ഐക്യരൂപം സാങ്കേതികവിദ്യയിൽ ഉപയോഗിക്കുന്നതിനായി സാങ്കേതികവിദ്യയെ പരിശോധിക്കുക.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ഐക്യരൂപം സാങ്കേതികവിദ്യയിൽ ഉപയോഗിക്കുന്നതിനായി സാങ്കേതികവിദ്യയെ പരിശോധിക്കുക. ഐക്യരൂപം സാങ്കേതികവിദ്യയിൽ ഉപയോഗിക്കുന്നതിനായി സാങ്കേതികവിദ്യയെ പരിശോധിക്കുക.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Electric Vehicle Hardware Lab.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.